Homework 01

Least squares, likelihood, optimization, collinearity, data preparation, and selection bias · August 31, 2026

Computational notes. All work in **Python 3** (NumPy 2.4, SciPy 1.17, pandas 3.0), with random numbers from NumPy's `default_rng(seed)` (PCG64). *Pseudo-random streams are language-specific, so numerical values differ from an R solution using* `set.seed()` *with the same seed, though every method, formula, and conclusion is identical.* Generation order within each question: predictor matrix first, then errors. All fits use `numpy.linalg.lstsq` (SVD via LAPACK `gelsd`); the normal equations are never formed and $(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$ is never computed. Question 1 is cross-checked against a Householder QR solve — the two agree to 1.3×10^{-14}. The error SD is read as $\varepsilon_i \sim N(0, 0.7^2)$, i.e. $\sigma^2 = 0.49$.

1 Question 1 Least squares and matrix calculations

```
rng  = np.random.default_rng(43201)
n, p = 100, 5
Xp   = rng.standard_normal((n, p))                # x_ij ~ iid N(0,1)
eps  = 0.7 * rng.standard_normal(n)               # eps_i ~ iid N(0, 0.7^2)
X    = np.column_stack([np.ones(n), Xp])          # [1  x1 ... x5]
beta = np.array([3.0, 1.4, -0.9, 0.7, 0.0, 1.1]);  y = X @ beta + eps
bhat, *_ = np.linalg.lstsq(X, y, rcond=None)      # SVD-based; no (X'X)^{-1}
```

(a) Dimensions, rank, and the estimate

$\boldsymbol{X} \in \mathbb{R}^{100\times6}$ with $\text{rank}(\boldsymbol{X}) = 6$. The rank equals the number of columns, so $\boldsymbol{X}$ has *full column rank*: columns are linearly independent, $\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite, and the solution is unique. Expected, since the five predictor columns are independent Gaussian draws (so linearly independent with probability one) and $n = 100 \gg 6$.

	intercept	x_1	x_2	x_3	x_4	x_5
True β_j	3.000	1.400	−0.900	0.700	0.000	1.100
$\hat\beta_j$	2.9731	1.3240	−0.8415	0.7204	−0.0831	1.1651
$\hat\beta_j - \beta_j$	−0.0269	−0.0760	0.0585	0.0204	−0.0831	0.0651

Comparison. Every estimate is close to the generating value; the largest deviation is 0.0831. They are not exactly equal, and should not be, since $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\varepsilon}$ carries the noise realization: the estimator is unbiased but has sampling variability of order $\sigma/\sqrt{n} = 0.07$. All six deviations lie within roughly one such standard error. Note $\hat\beta_4 = -0.0831$ where truly $\beta_4 = 0$: least squares does *not* return zero for an irrelevant predictor, only a small number reflecting chance correlation with the noise. Calling a coefficient "zero" needs a standard error, not a point estimate.

(b) Fitted values, residuals, RMSE

$\boldsymbol{r}^\top\boldsymbol{r} = 41.7755$, so RMSE $= \sqrt{41.7755/100} = \mathbf{0.6463}$. First five fitted values: 0.4284, 2.6798, 6.6011, −1.9718, 2.2709; residuals 1.0710, 0.2404, 0.0798, −0.0970, 0.4234. The residual mean is -2.2×10^{-16} (zero to machine precision), forced by the intercept.

The training RMSE sits slightly *below* the true $\sigma = 0.7$. This is expected: it divides by n rather than the degrees of freedom $n - 6 = 94$, and the fit has already used six directions to shrink the residuals. The corrected value is $\sqrt{41.7755/94} = 0.6666$.

(c) The quantity $\|\boldsymbol{X}^\top\boldsymbol{r}\|_\infty$

$\|\boldsymbol{X}^\top\boldsymbol{r}\|_\infty = \mathbf{1.28\times10^{-12}}$.

Why near zero. Least squares minimizes $S(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2$, and $\nabla S = -2\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \mathbf{0} \iff \boldsymbol{X}^\top\boldsymbol{r} = \mathbf{0}$. The normal equations *are* the statement $\boldsymbol{X}^\top\boldsymbol{r} = \mathbf{0}$, so this norm is exactly the largest violation of first-order optimality — zero in exact arithmetic. The 10^{-12} is floating-point rounding: entries of $\boldsymbol{X}^\top\boldsymbol{X}$ are of order 133, so relative to $\|\boldsymbol{X}^\top\boldsymbol{y}\|_\infty \approx 305$ the violation is $\approx 4\times10^{-15}$, a few units in the last place of double precision. The solver converged properly.

What it says about $\boldsymbol{r}$. The residual vector is *orthogonal to every column of* $\boldsymbol{X}$. Geometrically $\hat{\boldsymbol{y}}$ is the orthogonal projection of $\boldsymbol{y}$ onto $\mathcal{C}(\boldsymbol{X})$ and $\boldsymbol{r}$ lies in $\mathcal{C}(\boldsymbol{X})^\perp$, of dimension $n - 6 = 94$. Hence (i) $\mathbf{1} \in \mathcal{C}(\boldsymbol{X})$ gives

$\sum_i r_i = 0$, verified above; (ii) residuals are uncorrelated with every predictor in-sample, so no re-weighting of existing predictors can reduce the RSS — all linear signal in $\boldsymbol{X}$ is extracted; (iii) $\|\boldsymbol{y}\|^2 = \|\hat{\boldsymbol{y}}\|^2 + \|\boldsymbol{r}\|^2$, the ANOVA decomposition. This orthogonality is an algebraic identity holding for *any* $\boldsymbol{y}$, correct model or not. A small $\|\boldsymbol{X}^\mathsf{T}\boldsymbol{r}\|_\infty$ shows the linear algebra was done right; it is *not* evidence the linear model is correct. Nonlinearity would appear in a residual-versus-fitted plot, not here.

2 Question 2 Gaussian likelihood for linear regression

(a) Maximizing over $\boldsymbol{\beta}$ for fixed σ^2

$$\ell(\boldsymbol{\beta}, \sigma^2) = \underbrace{-\tfrac{n}{2}\log(2\pi\sigma^2)}_{\text{free of } \boldsymbol{\beta}} - \frac{1}{2\sigma^2}\underbrace{(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})}_{\mathrm{RSS}(\boldsymbol{\beta})} = c_1 - c_2\,\mathrm{RSS}(\boldsymbol{\beta}), \quad c_2 = \tfrac{1}{2\sigma^2} > 0.$$

The first term is an additive constant in $\boldsymbol{\beta}$, and c_2 is a strictly positive constant, so ℓ is a strictly *decreasing* affine function of $\mathrm{RSS}(\boldsymbol{\beta})$. A strictly decreasing transformation reverses optimization: maximizing ℓ over $\boldsymbol{\beta}$ is exactly minimizing $\mathrm{RSS}(\boldsymbol{\beta})$, with an identical argmax/argmin set.

Implication. The MLE of $\boldsymbol{\beta}$ under Gaussian errors *is* the OLS estimate, $\hat{\boldsymbol{\beta}}_{\mathrm{MLE}} = \hat{\boldsymbol{\beta}}_{\mathrm{OLS}}$, unique here by full column rank. The answer does not depend on σ^2 — the same $\boldsymbol{\beta}$ maximizes for every $\sigma^2 > 0$ — which is what makes the joint maximization separable, so we can solve for $\hat{\boldsymbol{\beta}}$ first and substitute it when estimating σ^2. Least squares needs no distributional assumption to be *defined*, but it acquires a likelihood justification precisely when errors are normal with constant variance; under heavier tails the MLE would no longer be OLS.

(b) MLE of σ^2

Fix $\boldsymbol{\beta}$, write $s = \mathrm{RSS}(\boldsymbol{\beta})$, $v = \sigma^2$:

$$\ell = -\tfrac{n}{2}\log(2\pi) - \tfrac{n}{2}\log v - \frac{s}{2v}, \qquad \frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{s}{2v^2} = \frac{1}{2v^2}(s - nv) \implies \boxed{\hat{\sigma}^2_{\mathrm{MLE}} = \mathrm{RSS}(\boldsymbol{\beta})/n}$$

This is a maximum: the derivative is positive for $v < s/n$ and negative for $v > s/n$ (equivalently $\partial^2\ell/\partial v^2 = -n^3/(2s^2) < 0$ there), and $\ell \to -\infty$ as $v \to 0^+$ and as $v \to \infty$, so it is the global maximum.

Estimator	Formula	$\hat{\sigma}^2$	$\hat{\sigma}$
Maximum likelihood	$\mathrm{RSS}/n = 41.7755/100$	0.41776	0.6463
Unbiased	$\mathrm{RSS}/(n-p) = 41.7755/94$	0.44442	0.6666
Truth	0.7^2	0.49000	0.7000

Here $p = 6$ estimated coefficients, so $n - p = 94$; $\hat{\sigma}_{\mathrm{MLE}} = 0.6463$ is numerically identical to the training RMSE of Q1(b) — the same formula.

Why the denominators differ. The ratio is exactly $\hat{\sigma}^2_{\mathrm{MLE}}/\hat{\sigma}^2_{\mathrm{unb}} = 94/100 = 0.94$, so the MLE is 6% too small on average. The reason is that RSS is evaluated at the *fitted* coefficients: least squares chooses $\hat{\boldsymbol{\beta}}$ to make residuals as small as possible, so $\mathrm{RSS}(\hat{\boldsymbol{\beta}}) \le \mathrm{RSS}(\boldsymbol{\beta})$ and the residuals are systematically optimistic. Formally $\boldsymbol{r} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{\varepsilon}$ with $\boldsymbol{H}$ the hat matrix, and since $\boldsymbol{I} - \boldsymbol{H}$ is an idempotent projection of rank $n - p$, $\mathbb{E}[\mathrm{RSS}] = \sigma^2\,\mathrm{tr}(\boldsymbol{I} - \boldsymbol{H}) = \sigma^2(n - p)$. Dividing by $n - p$ is exactly unbiased; dividing by n gives $\mathbb{E}[\hat{\sigma}^2_{\mathrm{MLE}}] = \sigma^2(n-p)/n < \sigma^2$. Geometrically, $\boldsymbol{r}$ is confined to the 94-dimensional space $\mathcal{C}(\boldsymbol{X})^\perp$ (Q1c), so RSS is a sum of 100 numbers carrying only 94 independent pieces of information; charging it 100 degrees of freedom double-counts. This is a general feature of maximum likelihood — consistent but not finite-sample unbiased — and it matters most when p is a large fraction of n.

(c) Profiling along the x_2 coefficient

With $\boldsymbol{e}_{x_2} = (0,0,1,0,0,0)^\top$ we evaluate $\ell(\hat{\boldsymbol{\beta}} + t\boldsymbol{e}_{x_2}, \hat{\sigma}^2_{\text{MLE}})$ on 1001 points $t \in [-1.25, 1.25]$. The maximum is at $\mathbf{t = 0}$, where $\ell = -98.2509$; at $t = \pm 1.25$ it has fallen to -347.827, a symmetric drop of 249.58 units.

Why this agrees with least squares. Since $\boldsymbol{X}(\hat{\boldsymbol{\beta}} + t\boldsymbol{e}_{x_2}) = \boldsymbol{X}\hat{\boldsymbol{\beta}} + t\boldsymbol{x}_2$,

$$\text{RSS}(\hat{\boldsymbol{\beta}} + t\boldsymbol{e}_{x_2}) = \|\boldsymbol{r} - t\boldsymbol{x}_2\|^2 = \|\boldsymbol{r}\|^2 - 2t\,\boldsymbol{x}_2^\top\boldsymbol{r} + t^2\boldsymbol{x}_2^\top\boldsymbol{x}_2,$$

and the cross term vanishes — this is precisely the orthogonality of Q1(c). Hence

$$\begin{aligned}\ell(\hat{\boldsymbol{\beta}} + t\boldsymbol{e}_{x_2}, \hat{\sigma}^2) &= \ell(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2) - \frac{\boldsymbol{x}_2^\top\boldsymbol{x}_2}{2\hat{\sigma}^2}\,t^2 \\ &= -98.2509 - 159.729\,t^2,\end{aligned}$$

using $\boldsymbol{x}_2^\top\boldsymbol{x}_2 = 133.455$ and $\hat{\sigma}^2 = 0.41776$. At $t = \pm 1.25$ this predicts -347.827, matching the grid to all printed digits.

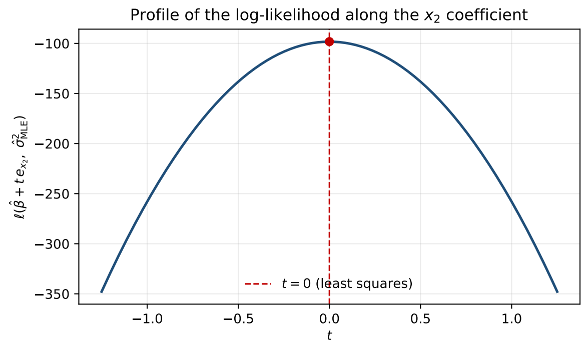

Figure 1: An exact downward parabola in t, peaking at $t = 0$.

So the curve is an exact downward parabola with vertex at $t = 0$ and *no linear term*. That absence is the whole point: the least-squares condition $\boldsymbol{x}_2^\top\boldsymbol{r} = 0$ is identical to the score condition $\partial\ell/\partial t|_{t=0} = 0$. Moving the x_2 coefficient away from $\hat{\beta}_2 = -0.8415$ can only raise the RSS and lower the likelihood — the one-dimensional slice of part (a). The curvature $-\boldsymbol{x}_2^\top\boldsymbol{x}_2/\hat{\sigma}^2$ is the observed Fisher information for this coordinate, corresponding to $\text{se}(\hat{\beta}_2) = 0.0589$; the grid spans about ± 21 standard errors, hence the dramatic endpoint drop. Note this is a *slice*, not a profile likelihood: the other five coefficients are held fixed rather than re-optimized, so the curvature reflects $(\boldsymbol{x}_2^\top\boldsymbol{x}_2)^{-1}$ rather than the $(2,2)$ entry of $(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$; the two coincide only if $\boldsymbol{x}_2$ is orthogonal to the other columns, which here it nearly but not exactly is.

3 Question 3 Starting values in nonconvex optimization

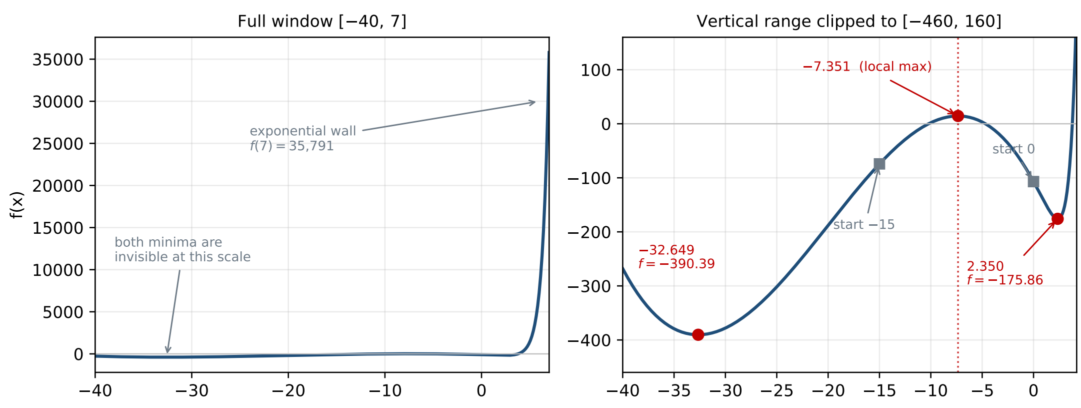

Figure 2: Left: f on the full window $[-40, 7]$; the exponential sets the entire vertical scale and both minima are invisible. Right: vertical range clipped, showing the two local minima (red), the local maximum at $x = -7.351$ separating their basins (dotted), and the two starting values (grey) on opposite sides of it.

(a) Features of the objective

1. **Nonconvex and multimodal.** Solving $f'(x) = 0$ gives exactly three stationary points on the real line:

$x^\star$	$f(x^\star)$	$f''(x^\star)$	Type
−32.6491	−390.3858	+3.795	local (and global) minimum
−7.3509	14.3858	−3.795	local maximum
2.3500	−175.8626	+69.687	local minimum

There are only three because for $x \lesssim -5$ the term $1.5e^{1.5x}$ is numerically negligible and $f'(x) \approx -0.15x^2 - 6x - 36$, a quadratic with roots −32.65 and −7.35; the third appears only where the exponential overtakes the quadratic. *Any local method can therefore return one of two answers, selected by the starting value.*

2. **The local maximum at −7.3509 is a watershed**, splitting the line into two basins of attraction: a descent method started to its left slides to −32.65, one started to its right slides to 2.35.
3. **Extreme scale disparity.** On $[-40, 7]$, f ranges from −390 to $f(7) = 35{,}791$, and essentially all variation is the exponential wall near the right endpoint; the two minima differ by only 214 units and are invisible on the full-range plot. Gradients are similarly mismatched ($|f'| \sim 10$–50 near the left minimum, $\sim 5 \times 10^4$ at $x = 7$), so no single step size or tolerance is well scaled for both regions.
4. **Overshoot/overflow risk on the right.** An over-long step right produces enormous function and gradient values (overflow beyond $x \approx 470$); robustness depends on the backtracking line search rejecting such steps.
5. **Bounded below, so a global minimum exists:** $e^{1.5x} \to \infty$ as $x \to +\infty$, and $-0.05x^3 \to +\infty$ dominates the quadratic as $x \to -\infty$. But the plotting window misleads on this point — f is still *decreasing* at the left endpoint ($f(-40) = -268$, above the minimum), so boundedness cannot be read off the plot.
6. **The left basin is shallow and nearly quadratic** (the exponential is negligible there), which is why BFGS converges in two iterations below.

(b) Two BFGS runs

```
f  = lambda x: np.exp(1.5*x) - 3*(x+6)**2 - 0.05*x**3
fp = lambda x: 1.5*np.exp(1.5*x) - 6*(x+6) - 0.15*x**2
for x0 in (-15.0, 0.0):
    res = minimize(lambda v: f(v[0]), np.array([x0]),
                   jac=lambda v: np.array([fp(v[0])]), method="BFGS")
```

Start $x^{(0)}$	Final x	Final $f(x)$	$\lvert f'(x)\rvert$	Iter./eval	Converged?	Optimizer message
−15	−32.64911	−390.38577	2.84×10^{-14}	2 / 6	Yes	*terminated successfully*
0	2.34997	−175.86262	8.13×10^{-6}	5 / 9	Yes	*terminated successfully*

Both report successful convergence, both gradients are numerically zero, both endpoints have positive curvature (3.795 and 69.687). *Both are genuine local minima and neither optimizer failed* — yet they differ by 214.5 units.

(c) Why they differ, and what globality would require

Why. BFGS is a strictly local descent method: it guarantees only a point with $f' = 0$, $f'' \geq 0$, and it never increases the objective, so it cannot climb an intervening hill. At $x^{(0)} = -15$, $f' = +20.25 > 0$, so the direction $-f'$ points left and, lying in the basin $(-\infty, -7.35)$, the iterates descend to −32.65. At $x^{(0)} = 0$, $f' = -34.5 < 0$, the direction points right, and the basin $(-7.35, \infty)$ leads to 2.35. The local maximum sits between the two starts as a barrier. Convexity is exactly the property that would rule this out; f'' changes sign, so the guarantee is lost. "Convergence reported" means only that the local first-order condition was met — *the success flag says nothing about globality.*

Which is lower. The run from −15: $f(-32.649) = -390.386$ versus $f(2.350) = -175.863$, lower by 214.52.

Additional evidence needed. (i) A *completeness argument for the stationary points* — solve $f'(x) = 0$ exhaustively by bracketing all sign changes on a fine grid and using Brent's method, giving the three roots above, plus the argument that f' is strictly signed for large $|x|$ so the list is complete; then compare f at the two minima. (ii) *Tail behaviour outside the window*, to ensure no minimum escapes to infinity (item 5 above) — this cannot be read off the plot. (iii) *Multistart*: BFGS from 200 starts spread over $[-40, 6]$ returns only

$\{-32.649,\ 2.350\}$ and nothing lower — corroboration, never proof, only a failure to find a counterexample. (iv) *Second-order confirmation*, $f''(-32.649) = 3.795 > 0$. In this 1-D analytically tractable problem (i)–(ii) amount to a proof; in higher dimensions such arguments are usually unavailable, which is why the honest statement is normally "the best local minimum found," and why reporting starting values is part of a reproducible result.

4 Question 4 Nearly collinear predictors

```
rng2 = np.random.default_rng(43202);  u = rng2.standard_normal(n)
x6   = X[:, 1] + 1e-4 * u             # x6 = x1 + 1e-4 u
Xa   = np.column_stack([X, x6])       # X+ = [X  x6];   ystar = y + 1e-3 * u
```

(a) Condition numbers and fits of $\boldsymbol{y}$

Design	Size	Condition number κ_2	Training RMSE
$\boldsymbol{X}$	100×6	1.4525	0.646340
$\boldsymbol{X}^+$	100×7	2.0340×10^4	0.633956

The condition number jumps four orders of magnitude. The mechanism shows in the singular values of $\boldsymbol{X}^+$: 13.606, 12.095, 11.163, 9.960, 9.947, 8.668, $\mathbf{6.689 \times 10^{-4}}$ — six of order 10, the seventh collapsed, because $\boldsymbol{x}_6 - \boldsymbol{x}_1 = 10^{-4}\boldsymbol{u}$ has norm only $\approx 10^{-3}$. The sample correlation between $\boldsymbol{x}_1$ and $\boldsymbol{x}_6$ is 0.999999995. The matrix still has full rank 7, so the solution is unique, but $\kappa_2 \approx 2 \times 10^4$ means about four decimal digits can be lost in the coefficients. The training RMSE improves only trivially (1.9%); that is not evidence $\boldsymbol{x}_6$ carries new signal, merely one extra degree of freedom absorbing a little noise along $\boldsymbol{u}$.

Coefficients in the augmented fit: $\hat{\beta}_{x_1} = -1329.743$, $\hat{\beta}_{x_6} = +1331.060$, sum $= 1.3171$. Individually these are absurd — three orders of magnitude too large, and $\hat{\beta}_{x_1}$ has flipped sign relative to the $\boldsymbol{X}$-only fit where $\hat{\beta}_{x_1} = 1.3240$. But they are enormous and *opposite*, and their sum is close to that stable value. The data determine the combined effect of the two nearly identical columns and say almost nothing about how to split it.

(b) Refitting with $\boldsymbol{y}^\star$

Coef.	Fit to $\boldsymbol{y}$	Fit to $\boldsymbol{y}^\star$	Change
$\hat{\beta}_{x_1}$	-1329.7430	-1339.7430	$\mathbf{-10.0000}$
$\hat{\beta}_{x_6}$	$+1331.0601$	$+1341.0601$	$\mathbf{+10.0000}$
Sum	1.31712	1.31712	2.3×10^{-13}

RMS difference of fitted values $= 9.7498 \times 10^{-4}$; maximum absolute difference $= 2.3512 \times 10^{-3}$. Training RMSE is 0.633956 in both fits, and all other coefficients are unchanged.

A response perturbation of size $\sim 10^{-3}$ moved each coefficient by 10 units — amplification of $\approx 10^4$, the order of the condition number — while moving fitted values by $\sim 10^{-3}$, i.e. not at all in practice.

(c) Explaining it via $\boldsymbol{y}^\star - \boldsymbol{y} = 10(\boldsymbol{x}_6 - \boldsymbol{x}_1)$

Since $\boldsymbol{x}_6 - \boldsymbol{x}_1 = 10^{-4}\boldsymbol{u}$, we have $\boldsymbol{y}^\star - \boldsymbol{y} = 10^{-3}\boldsymbol{u} = 10(\boldsymbol{x}_6 - \boldsymbol{x}_1)$. The key point: the perturbation is **exactly a linear combination of columns of $\boldsymbol{X}^+$**, namely $\boldsymbol{X}^+\boldsymbol{d}$ with $\boldsymbol{d} = 10(\boldsymbol{e}_{x_6} - \boldsymbol{e}_{x_1})$. It lies entirely in $\mathcal{C}(\boldsymbol{X}^+)$, with no component in $\mathcal{C}(\boldsymbol{X}^+)^\perp$.

Coefficients. Least squares is affine in the response, so adding something already in the column space shifts the solution exactly:

$$\hat{\boldsymbol{\beta}}^\star = \arg\min_{\boldsymbol{b}} \|\boldsymbol{y} + \boldsymbol{X}^+\boldsymbol{d} - \boldsymbol{X}^+\boldsymbol{b}\|^2 = \arg\min_{\boldsymbol{b}} \|\boldsymbol{y} - \boldsymbol{X}^+(\boldsymbol{b} - \boldsymbol{d})\|^2 = \hat{\boldsymbol{\beta}} + \boldsymbol{d}.$$

Hence $\hat{\beta}_{x_1}$ falls by exactly 10, $\hat{\beta}_{x_6}$ rises by exactly 10, all others are untouched, and the sum is invariant. Numerically $\hat{\boldsymbol{\beta}}^\star$ matches $\hat{\boldsymbol{\beta}} + \boldsymbol{d}$ to 6.8×10^{-13}, so the whole change is this identity, not numerical noise.

Fitted values. They change by $\boldsymbol{X}^+\boldsymbol{d} = 10(\boldsymbol{x}_6 - \boldsymbol{x}_1) = 10^{-3}\boldsymbol{u}$, with RMS $10^{-3}\,\mathrm{rms}(\boldsymbol{u}) = 9.75 \times 10^{-4}$ and maximum $10^{-3}\max|u_i| = 2.35 \times 10^{-3}$ — precisely the two numbers in part (b). The fitted values move by the size of the perturbation itself; nothing is amplified.

Why so different. $\|\boldsymbol{d}\|_2 = 10\sqrt{2} \approx 14.14$, but its image $\boldsymbol{X}^+\boldsymbol{d}$ has length only $\approx 9.7 \times 10^{-3}$. The direction

$\boldsymbol{e}_{x_6} - \boldsymbol{e}_{x_1}$ is very nearly in the null space of $\boldsymbol{X}^+$: moving far along it barely changes the fit, because it corresponds to the tiny singular value 6.689×10^{-4}. Fitted values live in the column space and depend on well-determined, large singular directions; the split of credit between $\boldsymbol{x}_1$ and $\boldsymbol{x}_6$ lives in the near-null direction, determined only by the microscopic $10^{-4}\boldsymbol{u}$. Sensitivity of $\hat{\boldsymbol{\beta}}$ scales like $1/\sigma_{\min} \approx 1.5 \times 10^3$, while that of the fitted values is bounded by 1.

Interpretation. Reporting separate "effects" for nearly collinear predictors is not meaningful, even though the software prints them without complaint and the fit is perfectly good. *Prediction is safe; interpretation is not* — both fits predict identically, yet a practitioner reading $\hat{\beta}_{x_1} = -1329.7$ would infer a huge *negative* effect when the generating value is $+1.4$. *Only estimable combinations should be reported*: the sum 1.3171 is stable to 10^{-13} and close to the true $\beta_1 = 1.4$; contrasts along well-conditioned directions are trustworthy, those along near-null directions are not. *Standard errors blow up too* (they scale with the square root of $[(\boldsymbol{X}^\top\boldsymbol{X})^{-1}]_{jj}$), so the coefficients would be reported as huge but insignificant — the model is saying it cannot separate them, and that message should be heeded. *Practical responses:* drop one redundant predictor, combine them, regularize (ridge penalizes the near-null direction), or report only the identified combination — diagnosing first via condition number, singular values, or VIFs. Finally, *this is not a numerical failure*: every fit used a stable SVD solver and each is the correct solution to the problem posed. The instability is statistical, not algorithmic; a better solver cannot recover information the data do not contain.

5 Question 5 Data preparation and summary statistics

(a) Reading, inspecting, building the analysis table

```
raw = pd.read_csv("data/data-manipulation.csv")
raw.shape;  raw.dtypes;  raw["observation_id"].duplicated().sum();  raw.isna().sum()
analysis = raw.loc[raw["response"].notna()].copy()      # no imputation with 0 or means
analysis["feature_sum"] = analysis["x1"] + analysis["x2"]
```

Dimensions: 24×5. **Column types:** `observation_id` and `group` are text; `x1`, `x2`, `response` are `float64`. **Duplicated identifiers: 0** (all 24 unique). **Missing values by column:** $0, 0, 0, 0, \mathbf{3}$ — all missingness confined to `response` (rows `obs-04`, `obs-11`, `obs-20`). Note `response` parses as `float64` rather than integer *because* the blanks become `NaN`, a useful sanity signal.

The **analysis table** retains $24 - 3 = \mathbf{21}$ observations and has 6 columns (the original five plus `feature_sum`). Following the instruction, the three missing responses are dropped, not imputed: imputing 0 would invent an extreme low response and drag every summary down, while imputing a group mean would leave group means unchanged but understate variability and fabricate precision. Dropping is the honest choice, though unbiased only if responses are missing at random — an assumption that cannot be checked from these data.

(b) Group summaries

Group	n	Mean response	Mean `feature_sum`
A	7	5.6000	3.6000
B	7	7.6029	4.9286
C	7	7.9971	6.5857
All	21	7.0667	5.0381

Which observations these describe. Only the 21 **observations with a recorded response** — not all 24 rows. The raw file has 8 observations in each of A, B, C; exactly one response is missing in each group, so the analysis table has 7 per group. The reported n is therefore a count of *complete-response* observations, not of enrolled observations. Here the loss happens to be balanced across groups, so the design balance survives — but that is luck, not guarantee, and worth checking rather than assuming. The `feature_sum` means are also computed on this reduced set even though `x1` and `x2` are fully observed for all 24 rows; one could compute them on all 24 and get different numbers, so both figures here are stated on the same 21 rows. Any conclusion (e.g. that C has the highest mean response) is conditional on missingness being unrelated to the response values.

(c) Top three responses and code checks

```
top3 = (analysis.sort_values("response", ascending=False)
              .head(3)[["observation_id", "group", "response", "feature_sum"]])
assert len(analysis) == raw["response"].notna().sum()   # row count is right
assert analysis["response"].isna().sum() == 0           # no missing response
assert analysis["observation_id"].is_unique             # identifiers unique
```

observation_id	group	response	feature_sum
obs-22	C	9.24	6.60
obs-16	B	9.00	5.20
obs-14	B	8.66	5.00

Check	Comparison	Result
Rows retained = nonmissing responses	21 = 21	**PASS**
No missing responses in analysis table	0 missing	**PASS**
Identifiers unique	21 of 21	**PASS**

The top three come from groups C and B, consistent with those having the higher group means, though `feature_sum` does not order the same way (6.60, 5.20, 5.00 against responses 9.24, 9.00, 8.66), so the response is not a deterministic function of the predictors. The checks are cheap and worth writing: the first catches filtering on the wrong column or dropping rows with missing *predictors* too, the second catches inverted filter logic, the third catches row duplication from a merge — one of the easier ways to corrupt a group mean silently. Using `assert` makes the pipeline *fail loudly* rather than produce plausible-looking but wrong summaries.

6 Question 6 Can 39 million Fitbit records represent US adults?

The researcher's claim confuses *sample size* with *representativeness*: sample size controls variance and does nothing about bias. The target is $\theta = \frac{1}{N}\sum_{i=1}^{N} \bar{s}_i$, the mean over all N US adults of each adult's own average daily step count, *with each adult weighted equally.*

(a) Are the 39 million records independent?

No. They are 39 million measurements on only 59,018 people — about 661 days each — with a clear two-level structure. *Within-person clustering:* daily counts vary around a person's own habitual level, so two days from the same participant are far more alike than two days from different participants, since they share occupation, commute, mobility, fitness, and neighbourhood. *Serial correlation:* consecutive days are correlated, with a weekday/weekend cycle, seasonal variation over 14 years, and persistence during illness, injury, or a change of job. The effective sample size is therefore far closer to the number of participants than to 39 million. Treating the records as independent understates standard errors by roughly the design effect $\sqrt{1 + (\bar{n} - 1)\rho}$, which with $\bar{n} \approx 661$ and even $\rho = 0.5$ is about 18.

What makes some contribute more days? *Enrollment timing* — the data span 14 years, so early enrollees contribute thousands of days and recent ones a few hundred, and BYOD participants may add retrospective history from before joining that WEAR participants by construction cannot. *Adherence and engagement* — wearing, charging, and syncing are voluntary, and people who enjoy self-tracking (often the more active) wear far more consistently. *Device continuity* — whether it breaks, is lost, or is abandoned once novelty fades; WEAR participants received theirs free and may have less attachment. *Health and life events* — illness, injury, hospitalization, caregiving, or a demanding job reduce recorded days, and these are exactly the periods when counts would be atypical. *Technology access* — syncing needs a smartphone and reliable internet.

How this weights participants unequally. With n_i recorded days and personal mean $\bar{s}_i$,

$$\hat{\theta}_{\text{naive}} = \frac{\sum_i \sum_j s_{ij}}{\sum_i n_i} = \sum_i \underbrace{\frac{n_i}{\sum_k n_k}}_{w_i} \bar{s}_i, \qquad \hat{\theta}_{\text{naive}} - \frac{1}{m}\sum_i \bar{s}_i = \frac{\operatorname{Cov}(n_i, \bar{s}_i)}{\bar{n}}.$$

This is a *weighted* mean, not the equally weighted one the target requires: a participant with 3,000 days counts about 30 times one with 100. The two agree only if recorded days are uncorrelated with activity level — implausible here, since the same enthusiasm, health, and routine that produce consistent wear also produce higher counts, so Cov > 0 is likely and the naive average is pulled toward high-adherence, high-activity participants. This particular problem is easy to fix (average within participants first), but that does not address (b) or (c).

(b) Does the BYOD–WEAR gap show that owning a Fitbit causes more walking?

No. This is observational: BYOD participants *already owned* a Fitbit, WEAR participants were *invited and given* one. Nobody was randomized to ownership, so the 6,867 − 5,797 = 1,070 step gap in medians is a difference between dissimilar populations, not a treatment effect. The study's own numbers document the imbalance — BYOD were far more likely to report being White (77.3% vs 55.1%) and far less likely to report household income of \$10,000–\$25,000 (6.1% vs 15.2%). Plausible alternatives:

1. **Confounding by socioeconomic status (most likely).** Buying a Fitbit requires disposable income and interest in fitness technology; the same income, education, occupation, and neighbourhood characteristics independently affect step counts through walkability, street safety, leisure time, job type (desk work vs manual labour), and chronic disease burden. Prior advantage is associated with both.
2. **Reverse causation / selection on the outcome.** People who already walk a lot are precisely the people who buy a step tracker. This alone could generate the entire gap.
3. **Differential measurement, not behaviour.** BYOD participants are habituated wearers; recipients of a free device may wear it sporadically or for fewer hours per day, which mechanically counts fewer steps. Part of the gap may lie in *how the days were recorded*, not in how much people walked.
4. **Health status at entry** may differ between someone recruited for health reasons and someone who bought a tracker recreationally.

At most, ownership is *associated* with higher recorded counts in this cohort. A causal claim would need randomization of device provision (WEAR is closer in spirit, but invitation and acceptance were still not random), or at minimum adjustment for measured confounders, comparable wear-time windows, and sensitivity analysis for unmeasured confounding.

(c) How selection can bias a very large dataset

(i) Selection of participants. A US adult appears only after a long chain of non-random filters: volunteering for All of Us (a volunteer cohort, not a probability sample); consenting to share wearable data; then either *already owning a Fitbit* — requiring disposable income, technology access, and interest in self-tracking — or being *invited to and accepting* WEAR. Sustaining participation additionally needs a compatible smartphone and reliable internet. The reported figures confirm the skew: 77.3% of BYOD reported being White against roughly 60% of US adults. Systematically under-represented are people with low income, without smartphones, older adults, rural residents, those with conditions limiting ambulation, and non-English speakers — groups whose step counts differ substantially. WEAR was evidently designed to counteract this (55.1% White, 15.2% lower income), a real improvement, but an invited-and-accepting subgroup is still not a probability sample with known inclusion probabilities.

(ii) Selection of recorded days. Even for an enrolled participant, a day enters only if the device was worn, charged, synced, and survived any wear-time quality filter. Missing days are not missing at random: disproportionately days of illness, injury, travel, unusual routine, disengagement, or a broken device. Partially worn days may record artificially low counts, and filtering them is itself a selection step. Recorded days therefore over-represent ordinary, engaged, healthy days. That the average participant contributes ≈ 661 days out of a possible $\sim 5{,}100$ shows how much of the calendar is absent.

Why the naive average fails. It is wrong on three counts at once: *wrong weighting* (by recorded days rather than equally, part a); *wrong population* (a self-selected subgroup, not US adults); and *wrong days* (a selected subset of each participant's days). Crucially, **none of these is repaired by more data** — sample size enters the variance, not the bias:

$$\hat{\theta}_{\text{naive}} - \theta \approx \underbrace{\rho_{R,S}}_{\text{data defect correlation}} \times \underbrace{\sqrt{(1-f)/f}}_{\text{data scarcity}} \times \underbrace{\sigma_S}_{\text{problem difficulty}},$$

where f is the sampled fraction and $\rho_{R,S}$ the correlation between inclusion and step count. With 59,018 participants out of roughly 258 million US adults, $f \approx 2.3 \times 10^{-4}$ and $\sqrt{(1-f)/f} \approx 66$, so even a tiny $\rho = 0.01$ gives an error of about $0.66\sigma_S$ — if the between-person SD of average daily steps is near 3,000, that is a bias around 2,000 steps, larger than the entire BYOD–WEAR gap.

This is the *big data paradox*: the standard error shrinks with n while the bias does not, so a huge selected dataset yields a very narrow confidence interval centred in the wrong place. More data of the same kind makes the answer look *more* certain and no more correct — arguably worse than a small random sample, which at least reports its uncertainty honestly. The "therefore" is a non sequitur: 39 million records buy precision, and precision is not accuracy.

What would help. Average within participants first; weight participants to US adult demographics by post-stratification, raking, or inverse-probability weighting against a benchmark such as NHANES; model the day-level missingness rather than ignoring it; report design-based uncertainty reflecting clustering; and conduct sensitivity analysis over plausible $\rho_{R,S}$. Even then the estimate rests on selection being ignorable given the adjustment variables — an assumption, not a fact. The direction of the residual bias is not determined here, and is not required.